

A REALIZATION OF F_4 FOR LITT PROBLEM NO. 13

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ABSTRACT. Let F be the Fano surface of a smooth cubic threefold with an Eckardt point, and let $E \subset F$ parametrize the lines through that point. The image X of F in $\text{Alb}(F)/E$ is an integral surface whose full convolution Tannaka group is F_4 on its minimal 26-dimensional module. The quotient contracts precisely E and splits the Albert module as $27 = 1 \oplus 26$. The fixed vector has nonzero Albert norm, as follows from the Hessian ranks on the singular cubic and connected irreducibility. The construction gives a seven-dimensional family of polarized ambient fourfolds. Its intersection complexes have characteristic cycle $\Lambda_X + 2\Lambda_0$ and generic twisted Hodge multiplicities $(6, 14, 6)$.

1. INTRODUCTION

For an integral subvariety Z of a complex abelian variety, the intersection complex IC_Z generates a Tannakian category under convolution, after objects of Euler characteristic zero are discarded. Litt Problem No. 13 asks whether the simple types G_2 , E_7 , E_8 , and F_4 occur as the resulting groups [10]. The construction below realizes F_4 by singular surfaces.

An Eckardt point of a smooth cubic threefold is a point whose tangent hyperplane section is a cone over a smooth plane cubic. Let $Y \subset \mathbf{P}^4$ be such a threefold, with Eckardt point p . Its Fano surface $F = F(Y)$ contains the elliptic curve E of lines through p . Choose an Albanese embedding $F \hookrightarrow A = \text{Alb}(F)$ based at a point of E , so that $E \subset A$ is an abelian subvariety, and put

$$q : A \longrightarrow B = A/E, \quad X = q(F)_{\text{red}}, \quad f = q|_F.$$

Theorem 1.1. *For every smooth cubic threefold with a marked Eckardt point, B is an abelian fourfold and $X \subset B$ is an integral symmetric surface with trivial translation stabilizer. The map f is birational, its only positive-dimensional fibre is $f^{-1}(0) = E$, and*

$$Rf_* \mathbf{C}_F[2] \simeq \text{IC}_X \oplus \delta_0, \quad \chi(\text{IC}_X) = 26.$$

The full convolution Tannaka pair is

$$(G(\text{IC}_X), \omega(\text{IC}_X)) \simeq (F_4, V_{26}),$$

where V_{26} is the minimal representation of F_4 .

The proof uses the E_6 Tannaka group of the Fano surface. Quotient pushforward gives $H = G(\text{IC}_X) \subset E_6$ and a restriction of the Albert module $J|_H = 1 \oplus V_{26}$. Triviality of the translation stabilizer makes V_{26} irreducible on H^0 . The Hessian

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of the Albert norm then forces the fixed line to contain a norm-one vector. Its full stabilizer is F_4 , and the classification of irreducible subgroups of the minimal module gives $H = F_4$.

2. CONVOLUTION AND THE NORMALIZED FANO SURFACE

All varieties are over $\mathbf{C}$. Intersection complexes have the perverse normalization: if $j : Z_{\text{reg}} \hookrightarrow Z$ and $d = \dim Z$, then $\text{IC}_Z = j_{!*}\mathbf{C}_{Z_{\text{reg}}}[d]$, extended by zero to the ambient abelian variety. Write δ_0 for the skyscraper sheaf at the origin. For addition m the convolution product is $K * L = Rm_*(K \boxtimes L)$.

The quotient by negligible objects is Tannakian on every rigid subcategory generated by a semisimple perverse sheaf of geometric origin [8, 5]. Its group is reductive, and $\dim \omega(K) = \chi(K)$. A quotient homomorphism $q : A \twoheadrightarrow B$ induces a tensor functor and a closed immersion

$$(2.1) \quad G(Rq_*K) \hookrightarrow G(K)$$

[8, Lemma 13.7]. Groups refer throughout to the full group; E_6 denotes its simply connected form.

The Albanese variety of F has dimension five, and its Albanese maps are embeddings [1]. For $s \in F$ let $L_s \subset Y$ be its line. Roulleau constructs an involution $\sigma = \sigma_E$ fixing E pointwise and a map $\gamma_E : F \rightarrow E$ such that, for general $s \notin E$,

$$(2.2) \quad Y \cap \langle p, L_s \rangle = L_s + L_{\gamma_E(s)} + L_{\sigma(s)}.$$

The induced Albanese automorphism has eigenvalue $+1$ on T_0E and -1 on T_0A/T_0E [11, Section 3.1 and Theorem 13]. Thus $q(\sigma(s)) = -q(s)$ and $X = -X$.

Lemma 2.1. *The Albanese embedding can be chosen with $E \subset A$ an abelian subvariety, the sum of every coplanar triple equal to zero, and $G(\text{IC}_F) = E_6$ on the Albert module J .*

Proof. For an embedding based at $e \in E$, addition is constant on the incidence variety $\mathcal{T} \subset F^3$ of ordered triples forming a plane section [1, 7]; call the value $c \in A$. By (2.2), its image in B is $x + 0 - x = 0$, so $c \in E$. Choose $u \in E$ with $3u = c$ and translate the embedding by $-u$. This preserves E and the quotient and makes the triple sum zero.

The incidence correspondence now gives a nonzero invariant symmetric cubic on J [7, Example 2.2(c)]. The derived connected Tannaka group and its representation are (E_6, J) [6, Theorem 2]. On this module the invariant cubic is a nonzero multiple of the Albert norm, whose full linear stabilizer is E_6 [7, Theorem 2.1(c)]. Faithfulness therefore gives $E_6 \subset G(\text{IC}_F) \subset E_6$. $\square$

3. THE QUOTIENT GEOMETRY

The tangent bundle theorem identifies

$$(3.1) \quad \mathbf{P}(T_s F) = L_s \subset \mathbf{P}(T_0 A) = \mathbf{P}^4, \quad \mathbf{P}(T_0 E) = \{p\}$$

[1][11, Proposition 4].

Proposition 3.1. *The map $f : F \rightarrow X$ is birational and immersive on $F \setminus E$. Its only contracted curve is E , and $f^{-1}(0) = E$.*

Proof. By (3.1), $\ker(df_s) \neq 0$ exactly when $p \in L_s$, that is, when $s \in E$. Any contracted curve consequently lies in E . Conversely, E is contracted, and $q^{-1}(0) = E$ in A gives the stated fibre. Thus f is generically finite onto an integral surface.

Choose a dense open $U \subset X_{\text{reg}}$ where f is finite and unramified, avoiding $B[2]$ and the nongeneric locus of (2.2). For $s, t \in f^{-1}(x)$ with $x \in U$, projection from p sends both L_s and L_t to $\mathbf{P}(T_x X)$. Hence both lie in $\Pi = \langle p, L_s \rangle$. The plane section (2.2) has exactly three lines. Its line through p maps to zero, so $t = s$ or $t = \sigma(s)$. The latter implies $x = -x$, contrary to $x \notin B[2]$. The generic degree is therefore one. $\square$

Proposition 3.2. *One has $Rf_* \mathbf{C}_F[2] \simeq \text{IC}_X \oplus \delta_0$ and $\chi(\text{IC}_X) = 26$.*

Proof. The proper birational surface map f is semismall. Its only fibre of positive dimension is the irreducible curve E over zero. Roulleau computes $E^2 = -3$ [11, Proposition 10]. The local intersection form is the nondegenerate matrix (-3) , so the semismall decomposition has exactly one skyscraper summand [3, Theorem 3.4.1]. Since $e(F) = 27$ [1], the Euler characteristic of the remaining summand is $27 - 1 = 26$. $\square$

Proposition 3.3. *$\text{Stab}_B(X) = \{0\}$, and the tautological module of $G(\text{IC}_X)$ is irreducible on the identity component.*

Proof. Factor f through the normalization $\nu : X^\nu \rightarrow X$, with $h : F \rightarrow X^\nu$. The only exceptional curve of h is E . A proper birational morphism of normal surfaces is an isomorphism outside its exceptional locus, so $X^\nu \setminus \{z\} \simeq F \setminus E$, where $z = h(E)$. The point z is singular: a birational morphism to a smooth surface factors locally into point blowups, whose exceptional curves are rational, whereas E is elliptic. Thus $\text{Sing}(X^\nu) = \{z\}$.

A translation preserving X lifts to its normalization and fixes z . Since $\nu(z) = 0$, the translation is zero. This is precisely nondivisibility of the simple, non-negligible sheaf IC_X . Connected irreducibility follows from [5, Proposition 3.3(2)]. $\square$

4. THE ALBERT NORM AND THE FULL GROUP

Put $H = G(\text{IC}_X)$ and $V = \omega(\text{IC}_X)$. By Lemma 2.1 and Proposition 3.2 and (2.1),

$$(4.1) \quad H \hookrightarrow E_6, \quad J|_H \simeq 1 \oplus V, \quad \dim V = 26.$$

Indeed the tensor unit already belongs to $\langle \text{IC}_X \rangle$, so $\langle \text{IC}_X \oplus \delta_0 \rangle = \langle \text{IC}_X \rangle$. Let $v_0 \neq 0$ span the trivial summand, and let N be the Albert cubic. Its complete polarization defines $\beta_v : J \rightarrow J^\vee$ by $\beta_v(u) = N(v, u, -)$.

Lemma 4.1. *For $v \neq 0$ with $N(v) = 0$, the kernel of β_v has dimension 17 in Albert rank one and 9 in Albert rank two.*

Proof. In octonion coordinates the cubic is

$$N = abc - a n(\xi) - b n(\eta) - c n(\zeta) + 2 \text{Re}((\xi\eta)\zeta),$$

where each octonion norm is nondegenerate of rank eight. The rank-orbit description of the Albert algebra reduces the two cases to $e_1 = \text{diag}(1, 0, 0)$ and $e_1 + e_2$ [13]. The quadratic terms at these points are, respectively,

$$bc - n(\xi), \quad \gamma(\alpha + \beta) - n(\xi) - n(\eta).$$

Their ranks are 10 and 18, giving kernel dimensions $27 - 10$ and $27 - 18$. $\square$

Proposition 4.2. *The fixed vector satisfies $N(v_0) \neq 0$.*

Proof. The map β_{v_0} is H^0 -equivariant. By Proposition 3.3, $J|_{H^0} = 1 \oplus V$ is a multiplicity-free sum of two irreducibles. Reductivity therefore gives

$$\dim \ker \beta_{v_0} \in \{0, 1, 26, 27\}.$$

If $N(v_0) = 0$, the vector has Albert rank one or two, contradicting Lemma 4.1. $\square$

Scale v_0 to norm one. Its full stabilizer in E_6 is the connected group F_4 [13]. Hence $H \subset F_4$, and V is its minimal module. The group H^0 is positive-dimensional, since it acts irreducibly on a space of dimension 26. No proper positive-dimensional closed subgroup of F_4 in characteristic zero acts irreducibly on this module: this is the F_4 row of [9, Theorem 1 and Table 1.2]. Thus

$$F_4 = H^0 \subset H \subset F_4.$$

This proves Theorem 1.1, including connectedness of the full group.

5. THE FAMILY AND ITS NUMERICAL INVARIANTS

Corollary 5.1. *The construction yields a seven-dimensional family of polarized ambient fourfolds of type $(1, 2, 2, 2)$. Every member carries the surface of Theorem 1.1.*

Proof. For a smooth cubic surface $S = \{g = 0\} \subset \mathbf{P}^3$ and a transverse hyperplane $\{\ell = 0\}$, the cubic $Y = \{g + x_0^2 \ell = 0\} \subset \mathbf{P}^4$ is smooth and has Eckardt point $[1 : 0 : 0 : 0 : 0]$. Let $P = \text{Im}(1 - \sigma) \subset J(Y) = A$. The invariant part is E , and the principal polarization identifies A/E with $P^\vee$. The polarization on P has type $(1, 1, 1, 2)$; its dual has type $(1, 2, 2, 2)$ [2, Lemma 1.12 and Remark 1.14]. The moduli space of these cubic surface pairs has dimension seven, and its period map to polarized P is injective with injective differential [2, Theorem 0.1]. Dualization gives the asserted dimension for the family of B . $\square$

Corollary 5.2 (A concrete example). *The cubic $x_0^2 x_1 + x_1^3 + x_2^3 + x_3^3 + x_4^3 = 0$, marked at $p = [1 : 0 : 0 : 0 : 0]$, gives a realization of (F_4, V_{26}) .*

Proof. The partial derivatives are $2x_0 x_1$, $x_0^2 + 3x_1^2$, and $3x_i^2$ for $i = 2, 3, 4$; they have no common projective zero. The tangent section at p is the cone $x_2^3 + x_3^3 + x_4^3 = 0$ in $x_1 = 0$. Apply Theorem 1.1. $\square$

For a subvariety $Z \subset B$, write Λ_Z for the closure of the conormal bundle to Z_{reg} in T^*B .

Proposition 5.3. *The characteristic cycle and conormal Gauss degree are*

$$\text{CC}(\text{IC}_X) = \Lambda_X + 2\Lambda_0, \quad \deg(\Lambda_X \rightarrow T_0^*B) = 24.$$

Proof. Over $B \setminus \{0\}$, f is finite and immersive. Locally its image is a union of smooth branches, and its direct image is the sum of their constant intersection complexes. By Proposition 3.2, no other conormal component occurs away from zero. Thus $\text{CC}(\text{IC}_X) = \Lambda_X + m\Lambda_0$.

A general covector in $T_0^*B \subset T_0^*A$ defines a general hyperplane $H \subset \mathbf{P}^4$ through p . By (3.1), a conormal point away from zero is a line $L_s \subset Y \cap H$ not passing through p . The surface $Y \cap H$ is a smooth cubic surface with Eckardt point p . Of its 27 lines, exactly three pass through p . The remaining 24 give the generic Gauss

fibre; exceptional strata are avoided by a general covector. Hence the degree is 24. The index formula on abelian varieties [4, Theorem 1.3] now gives $26 = 24 + m$. $\square$

Proposition 5.4. *For a general unitary character α of B , the twisted hypercohomology $H^0(B, \mathrm{IC}_X \otimes \mathcal{L}_\alpha)$ is pure of weight two with Hodge multiplicities $(6, 14, 6)$.*

Proof. Generic vanishing and Proposition 3.2 give $H^i(F, f^* \mathcal{L}_\alpha) = 0$ for $i \neq 2$ and dimension 27 in degree two [8, Theorem 1.1]. The unitary Hodge decomposition therefore makes the higher cohomology of $K_F \otimes f^* \alpha$ vanish. Riemann–Roch gives $h^0(K_F \otimes f^* \alpha) = \chi(K_F) = 6$, since $p_g(F) = 10$ and $q(F) = 5$ [1]. The same calculation for α^{-1} and duality give Hodge multiplicities $(6, 15, 6)$ for the twisted degree-two cohomology of F .

The rational Hodge module form of the semismall decomposition is $Rf_* \mathbf{Q}_F^H[2] = \mathrm{IC}_X^H \oplus \mathbf{Q}_0^H(-1)$ [3, 12]. The last term has type $(1, 1)$, yielding $(6, 14, 6)$ after its removal. $\square$

Remark 5.5. The 24 short roots and the zero weight of multiplicity two in V_{26} agree with the two terms of the characteristic cycle. The numerical calculation is independent of the subgroup argument.

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